COS 202 ASSIGNMENT

Name: Olorunfemi Emmanuel Adeleke
Matric No.: 241201061
Department: Computer Science

Course Code: COS 202

Solution 1

input

```
# Accept a three-digit integer from the user
num = int(input("Enter a three-digit integer: "))

# Input validation to ensure it's a valid three-digit number
if num < 100 or num > 999:
   print("Invalid input! Please enter a number between 100 and 999.")
else:
   # Extract digits mathematically without converting to a string
   hundreds = num // 100
   tens = (num // 10) % 10
   ones = num % 10

   # Determine the relationship between the digits
   if hundreds < tens and tens < ones:
      print(f"{num} (strictly increasing)")
   elif hundreds > tens and tens > ones:
      print(f"{num} (strictly decreasing)")
   elif hundreds == tens and tens == ones:
      print(f"{num} (all equal)")
   else:
      print(f"{num} (neither increasing nor decreasing)")
```

output

```
Enter a three-digit integer:
```

Solution 2

input

```
# Initial setup
balance = 75000
```

```
print("--- Simple ATM ---")
print("1. Deposit")
print("2. Withdraw")
print("3. Transfer")
print("4. Balance Enquiry")
print("5. Exit")

choice = int(input("Enter your choice (1-5): "))

match choice:
    case 1:
        amount = float(input("Enter amount to deposit: ₦"))
        if amount > 0:
            balance += amount
            print(f"Success: ₦{amount:,.2f} deposited successfully.")
            print(f"Current Balance: ₦{balance:,.2f}")
        else:
            print("Error: Deposit amount must be greater than zero.")
    case 2:
        amount = float(input("Enter amount to withdraw: ₦"))
        if amount <= balance:
            balance -= amount
            print(f"Success: Please take your cash of ₦{amount:,.2f}.")
            print(f"Current Balance: ₦{balance:,.2f}")
        else:
            print("Error: Withdrawal amount exceeds available balance.")
    case 3:
        acct_num = input("Enter recipient's account number: ")
        amount = float(input("Enter transfer amount: ₦"))
        if amount <= balance:
            balance -= amount
            print(f"Success: ₦{amount:,.2f} transferred to account {acct_num}.")
            print(f"Current Balance: ₦{balance:,.2f}")
        else:
            print("Error: Transfer amount exceeds available balance.")
    case 4:
        print(f"Your current account balance is: ₦{balance:,.2f}")
    case 5:
        print("Thank you for using our ATM. Goodbye!")
    case _:
```

```
        print("Error: Invalid menu choice selected.")
```

<u>Output</u>

```
--- Simple ATM ---
1. Deposit
2. Withdraw
3. Transfer
4. Balance Enquiry
5. Exit
Enter your choice (1-5):
```

<u>Solution 3</u>

<u>Input</u>

```
jamb = int(input("Enter JAMB Score: "))
credits = int(input("Enter Number of O'Level Credits: "))
age = int(input("Enter Age: "))

if jamb >= 300 and credits >= 8 and age >= 16:
    print("Scholarship Candidate")
elif jamb >= 200 and credits >= 5 and age >= 16:
    print("Admission Offered")
else:
    print("Admission Denied")
    print("Reason(s) for denial:")
    if jamb < 200:
        print(f"- Your JAMB score ({jamb}) is below the required minimum of 200.")
    if credits < 5:
        print(f"- Your O'Level credits ({credits}) are below the required minimum of 5.")
    if age < 16:
        print(f"- Your age ({age}) is below the minimum entry age of 16.")
```

<u>Output</u>

```
Enter JAMB Score:
```

<u>Solution 4</u>

<u>input</u>

```
a = float(input("Enter side 1: "))
b = float(input("Enter side 2: "))
```

```
c = float(input("Enter side 3: "))

# Checking Triangle Inequality Theorem
if (a + b > c) and (a + c > b) and (b + c > a):
    if a == b == c:
        print("Equilateral Triangle")
    elif a == b or b == c or a == c:
        print("Isosceles Triangle")
    else:
        print("Scalene Triangle")
else:
    print("Invalid Triangle")
```

output

```
Enter side 1:
```

Solution 5

input

```
print("Customer Categories:\n1 - Regular\n2 - Silver\n3 - Gold")
category_id = int(input("Enter Customer Category (1-3): "))
purchase_amount = float(input("Enter Purchase Amount: ₦"))
day = input("Is the shopping day Friday? (Yes/No): ").strip().lower()

discount_pct = 0.0
category_name = ""

if category_id == 1:
    category_name = "Regular"
    if purchase_amount > 20000:
        discount_pct = 5.0
elif category_id == 2:
    category_name = "Silver"
    if 20001 <= purchase_amount <= 50000:
        discount_pct = 10.0
    elif purchase_amount > 50000:
        discount_pct = 15.0
elif category_id == 3:
    category_name = "Gold"
    discount_pct = 20.0
```

```
# Friday conditional check
if day == "yes":
    discount_pct += 5.0

# Caps upper limit threshold
if discount_pct > 25.0:
    discount_pct = 25.0

discount_value = (discount_pct / 100) * purchase_amount
final_payable = purchase_amount - discount_value

print("\n--- Receipt Summary ---")
print(f"Customer Category: {category_name}")
print(f"Purchase Amount: ₦{purchase_amount:,.2f}")
print(f"Total Discount (%): {discount_pct}%")
print(f"Discount Value: ₦{discount_value:,.2f}")
print(f"Final Amount Payable: ₦{final_payable:,.2f}")
```

<u>output</u>

```
Customer Categories:
1 - Regular
2 - Silver
3 - Gold
Enter Customer Category (1-3):
```

<u>solution 6</u>

<u>input</u>

```
n1 = int(input("Enter first integer: "))
n2 = int(input("Enter second integer: "))
n3 = int(input("Enter third integer: "))

if n1 < n2 and n1 < n3:
    if n2 < n3:
        print(f"Ascending order: {n1}, {n2}, {n3}")
    else:
        print(f"Ascending order: {n1}, {n3}, {n2}")
elif n2 < n1 and n2 < n3:
    if n1 < n3:
        print(f"Ascending order: {n2}, {n1}, {n3}")
    else:
```

```
        print(f"Ascending order: {n2}, {n3}, {n1}")
else:
    if n1 < n2:
        print(f"Ascending order: {n3}, {n1}, {n2}")
    else:
        print(f"Ascending order: {n3}, {n2}, {n1}")
```

<u>output</u>

Enter first integer:

<u>solution 7</u>

<u>input</u>

```
cond1 = input("Is the password correct? (Yes/No): ").strip().lower()
cond2 = input("Is the fingerprint verified? (Yes/No): ").strip().lower()
cond3 = input("Is face recognition successful? (Yes/No): ").strip().lower()

# Convert answers to Boolean integers (1 for True, 0 for False)
c1 = 1 if cond1 == "yes" else 0
c2 = 1 if cond2 == "yes" else 0
c3 = 1 if cond3 == "yes" else 0

if (c1 + c2 + c3) >= 2:
    print("Access Granted: The security door opens.")
else:
    print("Access Denied: Insufficient authorization matches.")
```

<u>output</u>

Is the password correct? (Yes/No):

<u>solution 8</u>

<u>input</u>

```
ca = float(input("Enter CA Score (0-30): "))
exam = float(input("Enter Examination Score (0-70): "))
attendance = float(input("Enter Attendance (%): "))

total_score = ca + exam

if attendance < 75:
    grade = "F"
```

```
else:
    if 70 <= total_score <= 100:
        grade = "A"
    elif 60 <= total_score <= 69:
        grade = "B"
    elif 50 <= total_score <= 59:
        grade = "C"
    elif 45 <= total_score <= 49:
        grade = "D"
    else:
        grade = "F"

print("\n--- Academic Result ---")
print(f"Total Score: {total_score}")
print(f"Grade: {grade}")
```

output

Enter CA Score (0-30):

solution 9

input

```
distance = float(input("Enter distance traveled (in km): "))
age = int(input("Enter passenger age: "))
is_student = input("Are you a student? (Yes/No): ").strip().lower()

# Gross Fare Assessment
base_fare = 1000
gross_fare = base_fare

if distance <= 10:
    pass
elif distance <= 30:
    gross_fare += (distance - 10) * 80
else:
    gross_fare += (20 * 80) + ((distance - 30) * 120)

# Discount Assessment
discount_pct = 0.0
if age > 60:
    discount_pct += 15.0
```

```
if is_student == "yes":
    discount_pct += 10.0

if discount_pct > 20.0:
    discount_pct = 20.0

discount_val = (discount_pct / 100) * gross_fare
final_fare = gross_fare - discount_val

print("\n--- Fare Breakdown ---")
print(f"Distance: {distance} km")
print(f"Gross Fare: ₦{gross_fare:,.2f}")
print(f"Discount Applied: {discount_pct}% (-₦{discount_val:,.2f})")
print(f"Final Fare: ₦{final_fare:,.2f}")
```

output

```
Enter distance traveled (in km):
```

solution 10

input

```
s1 = float(input("Enter first score: "))
s2 = float(input("Enter second score: "))
s3 = float(input("Enter third score: "))

avg = (s1 + s2 + s3) / 3

if avg >= 80:
    print(f"Average: {avg:.2f} -> Excellent Student")
elif avg >= 60:
    print(f"Average: {avg:.2f} -> Good Student")
elif avg >= 50:
    print(f"Average: {avg:.2f} -> Average Student")
else:
    print(f"Average: {avg:.2f} -> Probation Student")
```

output

```
Enter first score:
```

solution 11

input

```
age = int(input("Enter age: "))
citizenship = input("Are you a citizen? (Yes/No): ").strip().lower()
criminal_record = input("Do you have a prior criminal record? (Yes/No): ").strip().lower()

if citizenship != "yes":
    print("Output: Not Eligible (Non-citizens cannot vote)")
elif age < 18:
    print("Output: Not Eligible (Underage)")
elif criminal_record == "yes":
    print("Output: Special Verification Required (Requires legal clearance evaluation)")
else:
    print("Output: Eligible to Vote")
```

output

```
Enter age:
```

solution 12

input

```
# Executed using a continuous test loop structure
while True:
    print("\n--- Football Score Classifier (Type '-1' to exit) ---")
    t1_score = int(input("Enter Home Team Score: "))
    if t1_score == -1:
        break
    t2_score = int(input("Enter Away Team Score: "))

    diff = abs(t1_score - t2_score)

    if diff == 0:
        print("Result: Balanced Draw Match")
    elif diff == 1:
        print("Result: Narrow Victory")
    elif 2 <= diff <= 3:
        print("Result: Heavy Victory")
    else:
        print("Result: Absolute Thrashing!")
```

output
--- Football Score Classifier (Type '-1' to exit) ---
Enter Home Team Score:

solution 13
input
```
level = int(input("Enter Level (e.g., 100, 400): "))
cgpa = float(input("Enter CGPA (0.00 - 5.00): "))
has_fees = input("Do you have outstanding fees? (Yes/No): ").strip().lower()
reg_status = input("Is course registration open? (Yes/No): ").strip().lower()

# Primary Match-Case checking workflow dependencies
match (reg_status, has_fees):
    case ("no", _):
        print("Advisor Recommendation: Registration Closed")
    case (_, "yes"):
        print("Advisor Recommendation: Clear Outstanding Fees")
    case ("yes", "no"):
        if cgpa < 1.5:
            print("Advisor Recommendation: Academic Probation (Meet Your
Adviser)")
        elif level >= 400 and cgpa >= 1.5:
            print("Advisor Recommendation: Graduation Eligible")
        else:
            print("Advisor Recommendation: Registration Approved")
```

output
Enter Level (e.g., 100, 400):

solution 14
input
```
age = int(input("Enter Age: "))
income = float(input("Enter Monthly Income: ₦"))
employment_years = int(input("Enter Years of Employment: "))
existing_loan = input("Do you have an existing loan? (Yes/No): ").strip().lower()
credit_rating = input("Enter Credit Rating (Excellent, Good, Fair, Poor):
").strip().capitalize()

# Track individual rejection grounds via conditional statements
```

```
rejections = 0

if age < 21 or age > 60:
   print("Rejected: Age must be between 21 and 60 years.")
   rejections += 1
if income < 100000:
   print("Rejected: Monthly income must be at least ₦100,000.")
   rejections += 1
if employment_years < 2:
   print("Rejected: Minimum continuous employment required is 2 years.")
   rejections += 1
if credit_rating == "Poor":
   print("Rejected: Credit assessment history profile rated poor.")
   rejections += 1

if rejections == 0:
   if existing_loan == "yes" or credit_rating == "Fair":
      print("Status Decision: Loan Approved with Conditions")
   else:
      print("Status Decision: Loan Approved")
```

<u>output</u>

Enter Age:

<u>solution 15</u>

<u>input</u>

```
# Absolute Constraints Observed: No Loops, No Lists, No Functions. Only basic sequence blocks.
s1 = float(input("Enter Course 1 Score: "))
s2 = float(input("Enter Course 2 Score: "))
s3 = float(input("Enter Course 3 Score: "))
s4 = float(input("Enter Course 4 Score: "))
s5 = float(input("Enter Course 5 Score: "))

total = s1 + s2 + s3 + s4 + s5
average = total / 5

if average >= 70:
   classification = "First Class"
   status = "Pass"
```

```
elif average >= 60:
   classification = "Second Class Upper"
   status = "Pass"
elif average >= 50:
   classification = "Second Class Lower"
   status = "Pass"
elif average >= 45:
   classification = "Third Class"
   status = "Pass"
else:
   classification = "Fail"
   status = "Fail"

print("\n--- Academic Result Sheet Summary ---")
print("Total Marks Obtained:", total)
print("Average Mark Value  :", average)
print("Grade Classification:", classification)
print("Pass/Fail Status    :", status)
```

output

```
Enter Course 1 Score:
```

solution 16

input

```
# Restrictions: No loops, lists, functions, or external module extensions.
print("==================================================")
print("     SMART NIGERIAN CITIZEN SERVICE PORTAL    ")
print("==================================================")
print("1. WAEC Result Checker")
print("2. JAMB Admission Adviser")
print("3. Voter Eligibility Checker")
print("4. Loan Eligibility Checker")
print("5. Electricity Bill Calculator")
print("6. Exit")
print("==================================================")

choice = int(input("Select service portal tracking item option (1-6): "))

match choice:
   case 1:
```

```
        print("\n--- [WAEC Result Checker] ---")
        maths = input("Grade in Mathematics (A1-F9): ").upper()
        eng = input("Grade in English (A1-F9): ").upper()
        print(f"Status: Record verified. Core subjects checked ({maths}, {eng}).
Printing printable slip...")
    case 2:
        print("\n--- [JAMB Admission Adviser] ---")
        score = int(input("Enter your JAMB score: "))
        if score >= 200:
            print("Advice: You are eligible to apply for University admission paths.")
        else:
            print("Advice: Remedial options or technical colleges suggested.")
    case 3:
        print("\n--- [Voter Eligibility Checker] ---")
        v_age = int(input("Enter your current age: "))
        v_cit = input("Are you a Nigerian Citizen? (Yes/No): ").lower()
        if v_age >= 18 and v_cit == "yes":
            print("Status: Eligible. Proceed to PVC collection desk centers.")
        else:
            print("Status: Ineligible for national voters register database updates.")
    case 4:
        print("\n--- [Loan Eligibility Checker] ---")
        sal = float(input("Enter verifiable monthly salary income (₦): "))
        if sal >= 50000:
            print("Status: Provisionally pre-approved for standard credit facilities.")
        else:
            print("Status: Income minimum threshold criteria unmet.")
    case 5:
        print("\n--- [Electricity Bill Calculator] ---")
        units = float(input("Enter energy power units consumed (kWh): "))
        bill = units * 120 # Flat grid multiplier band rate
        print(f"Total Current NEPA/DisCo Tariff Charge Due: ₦{bill:,.2f}")
    case 6:
        print("\nThank you for using the Citizen Service Portal. Goodbye!")
    case _:
        print("\nInvalid operation channel index selected.")
```

output

```
==================================================
   SMART NIGERIAN CITIZEN SERVICE PORTAL
==================================================
```

1. WAEC Result Checker
2. JAMB Admission Adviser
3. Voter Eligibility Checker
4. Loan Eligibility Checker
5. Electricity Bill Calculator
6. Exit
==
Select service portal tracking item option (1-6):

solution 17

input

```
# Constraint: No strings allowed. Use math modulo operators on base inputs.
pin = int(input("Enter 4-digit PIN to evaluate: "))

if pin < 1000 or pin > 9999:
   print("Access Denied: PIN must have exactly 4 digits.")
else:
   d1 = pin // 1000
   d2 = (pin // 100) % 10
   d3 = (pin // 10) % 10
   d4 = pin % 10

   middle_pair = (d2 * 10) + d3
   digit_sum = d1 + d2 + d3 + d4
   ends_product = d1 * d4

   if (d1 + d4) % 2 != 0:
      print("Access Denied: The sum of the first and last digit is not even.")
   elif (middle_pair % 7 != 0) and (middle_pair % 13 != 0):
      print("Access Denied: Middle pair is not a multiple of 7 or 13.")
   elif digit_sum <= ends_product:
      print("Access Denied: Digit sum is not greater than the product of the outer
ends.")
   else:
      print("Access Granted")
```

output

Enter 4-digit PIN to evaluate:

solution 18

input

```
# Constraint: Use only if-elif-else blocks. Extract elements strictly via math.
secret = int(input("Set 3-digit secret code (000-999): "))
guess = int(input("Enter player guess code (000-999): "))

s1, s2, s3 = secret // 100, (secret // 10) % 10, secret % 10
g1, g2, g3 = guess // 100, (guess // 10) % 10, guess % 10

if secret == guess:
   print("The chest opens! You found treasure!")
else:
   # Evaluate matches
   match_pos1 = 1 if s1 == g1 else 0
   match_pos2 = 1 if s2 == g2 else 0
   match_pos3 = 1 if s3 == g3 else 0
   exact_matches = match_pos1 + match_pos2 + match_pos3

   # Check digit presence anywhere in the pool
   c1 = 1 if (g1 == s1 or g1 == s2 or g1 == s3) else 0
   c2 = 1 if (g2 == s1 or g2 == s2 or g2 == s3) else 0
   c3 = 1 if (g3 == s1 or g3 == s2 or g3 == s3) else 0
   total_found = c1 + c2 + c3

   if exact_matches == 1 and total_found == 1:
      print("One digit locked in place!")
   elif exact_matches == 0 and total_found == 1:
      print("One digit is correct but misplaced!")
   elif total_found == 2:
      print("Two digits are correct!")
   else:
      print("Nothing matches. The chest trembles...")
```

output

```
Set 3-digit secret code (000-999):
```

solution 19

input

```
t1 = input("Enter Ingredient 1 Type (Fire/Water/Earth/Air): ").strip().capitalize()
q1 = int(input("Enter Ingredient 1 Quantity (1-5): "))
```

```python
t2 = input("Enter Ingredient 2 Type (Fire/Water/Earth/Air): ").strip().capitalize()
q2 = int(input("Enter Ingredient 2 Quantity (1-5): "))

valid_types = ("Fire", "Water", "Earth", "Air")

if t1 not in valid_types or t2 not in valid_types:
    print("Invalid alchemy!")
else:
    # Use match-case tuple structural matching strategy
    match (t1, t2):
        case ("Fire", "Water") | ("Water", "Fire"):
            print("Steam Cloud" if q1 == q2 else "Sizzling Sludge")
        case ("Fire", "Earth") | ("Earth", "Fire"):
            print("Lava Bomb" if (q1 + q2) > 6 else "Magma Paste")
        case ("Fire", "Air") | ("Air", "Fire"):
            print("Explosion" if (q1 == 5 or q2 == 5) else "Smoke Screen")
        case ("Water", "Earth") | ("Earth", "Water"):
            print("Mud Golem" if (q1 % 2 == 0 and q2 % 2 == 0) else "Clay Ball")
        case ("Water", "Air") | ("Air", "Water"):
            print("Ice Shard" if abs(q1 - q2) == 1 else "Fog Mist")
        case ("Earth", "Air"):
            print("Dust Devil" if q1 > q2 else "Sand Trap")
        case ("Air", "Earth"):
            print("Dust Devil" if q2 > q1 else "Sand Trap")
        case ("Fire", "Fire"):
            print("Inferno" if (q1 + q2) > 6 else "Ember")
        case ("Water", "Water"):
            print("Torrent" if (q1 + q2) > 6 else "Puddle")
        case ("Earth", "Earth"):
            print("Boulder" if (q1 + q2) > 6 else "Pebble")
        case ("Air", "Air"):
            print("Gale" if (q1 + q2) > 6 else "Breeze")
```

<u>output</u>

```
^[[23~Enter Ingredient 1 Type (Fire/Water/Earth/Air):
```